Significant Does Not Mean Caused

Unit 3 · Topic 3.7 · TODAY'S PROBLEM

A cafeteria served 5,000 September trays after new sorting-bin labels were installed. The director claims at least 60% are sorted correctly. An SRS of 250 trays finds 136 correct; the club tests at $\alpha = 0.05$.

Imani: "The p-value is 0.0354, so reject H_0. The labels caused sorting below 60 percent for every month."

Drew: "The 5.6-point gap may be too small to matter, so it is not significant. Accept the director's claim."

September study

Source	Trays	Correct	α
September	5,000	claim: $\geq 60\%$	–
SRS	250	136	0.05

FIRST TAKE (5 min, on your sheet)

Which parts of Imani's and Drew's statements do you trust? Mark one statistical claim and one scope or importance claim to revisit.

DOK 1 **(a)** Define the population proportion p, state H_0 and H_a, and calculate $\widehat{p}$ and its signed difference from 0.60.

DOK 2 **(b)** Verify the Random, 10 percent, and Large Counts conditions. Calculate the one-proportion z statistic and lower-tail p-value.

DOK 3 **(c)** Adjudicate both statements and give the strongest conclusion at $\alpha = 0.05$. Interpret 0.0354 conditionally, distinguish significance from practical importance, bound cause and scope, and give the reader consequence of each overreach.

Video + follow-along: `u6_lesson6_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.